Assignment:

Module -1: Understanding of Hardware and Its Components

Section 1: Multiple Choice

1. Which of the following is NOT a component of the CPU?
    1. ALU
    2. RAM
    3. CU
    4. 1 and 3 both
2. What is the function of RAM in a computer?
3. Which of the following is a primary storage device?
    1. HDD
    2. SSD
    3. SD card
    4. 1 and 2 both
4. What is the purpose of a GPU?

Section 2: True or False

5. True or False: The motherboard is the main circuit board of a computer where other components are attached.
6. True or False: A UPS (Uninterruptible Power Supply) is a hardware device that provides emergency power to a load when the input power source fails.
7. True or False: An expansion card is a circuit board that enhances the functionality of a component.

Section 3: Short Answer

8. Explain the difference between HDD and SSD.
9. Describe the function of BIOS in a computer system.
10. List and briefly explain three input devices commonly used with computers.

Section 4: Practical Application

11. Identify and label the following components on a diagram of a motherboard:
    - CPU
    - RAM slots
    - SATA connectors

- PCI-E slot

12. Demonstrate how to install a RAM module into a computer.

Section 5: Essay

13. Discuss the importance of proper cooling mechanisms in a computer system. Include examples of cooling methods and their effectiveness.
14. Explain the concept of bus width and its significance in computer architecture.

Assignment module 2 : Installation and Maintenance of Hardware and Its

Section 1: Multiple Choice

1. Which of the following precautions should be taken before working on computer hardware?
   a) Ensure the computer is plugged in to prevent electrostatic discharge.
   b) Wear an anti-static wrist strap to prevent damage from electrostatic discharge.
   c) Work on carpeted surfaces to prevent slipping.
   d) Use magnetic tools to handle components more easily.

2. What is the purpose of thermal paste during CPU installation?
   a) To insulate the CPU from heat.
   b) To provide mechanical support for the CPU.
   c) To improve thermal conductivity between the CPU and the heat sink.
   d) To prevent the CPU from overheating.

3. Which tool is used to measure the output voltage of a power supply unit (PSU)?
   a) Multimeter
   b) Screwdriver
   c) Pliers
   d) Hex key

4. Which component is responsible for storing BIOS settings, such as date and time, even when the computer is powered off?
   a) CMOS battery
   b) CPU
   c) RAM
   d) Hard drive

Section 2: True or

5. True or False: When installing a new hard drive, it is essential to format it before use.
6. True or False: A POST (Power-On Self-Test) error indicates a problem with the CPU.
7. True or False: It is safe to remove a USB flash drive from a computer without ejecting it first.

Section 3: Short Answer

8. Describe the steps involved in installing a new graphics card in a desktop computer.
9. What is RAID, and what are some common RAID configurations?

Section 4: Practical Application

10. Demonstrate how to replace a CPU fan in a desktop computer.

Section 5: Essay

11. Discuss the importance of regular maintenance for computer hardware and provide examples of maintenance tasks.

Assignment module 3 : Understanding and Maintenance of

Section 1: Multiple Choice

1. What is the primary function of a router in a computer network?
   a) Assigning IP addresses to devices
   b) Providing wireless connectivity to devices
   c) Forwarding data packets between networks
   d) Managing user authentication and access control
2. What is the purpose of DNS (Domain Name System) in a computer network?
   a) Encrypting data transmissions for security
   b) Assigning IP addresses to devices dynamically
   c) Converting domain names to IP addresses

d) Routing data packets between network segments

3. What type of network topology uses a centralized hub or switch to connect all devices?
   a) Star
   b) Bus
   c) Ring
   d) Mesh
4. Which network protocol is commonly used for securely accessing and transferring files over a network?
   a) HTTP
   b) FTP
   c) SMTP
   d) POP3

Section 2: True or False

5. True or False: A firewall is a hardware or software-based security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.
6. True or False: DHCP (Dynamic Host Configuration Protocol) assigns static IP addresses to network devices automatically.
7. True or False: VLANs (Virtual Local Area Networks) enable network segmentation by dividing a single physical network into multiple logical networks.

Section 3: Short Answer

8. Explain the difference between a hub and a switch in a computer network.
9. Describe the process of troubleshooting network connectivity issues.

Section 4: Practical Application

10. Demonstrate how to configure a wireless router's security settings to enhance network security.

Section 5: Essay

11. Discuss the importance of network documentation and provide examples of information that should be documented.

Assignment module 4: Troubleshooting and

Section 1: Multiple Choice

1. What is the first step in the troubleshooting process?
   a) Implementing a solution
   b) Identifying the problem
   c) Testing the solution
   d) Documenting the solution
2. Which of the following tools is commonly used to diagnose hardware issues by testing electrical connections?
   a) Loopback plug
   b) Toner probe
   c) Multimeter
   d) Cable tester
3. Which Windows utility can be used to view system logs, monitor performance, and diagnose hardware and software issues?
   a) Task Manager
   b) Device Manager
   c) Event Viewer
   d) Control Panel

Section 2: True or False

4. True or False: Safe Mode is a diagnostic mode in Windows that loads only essential system services and drivers, allowing users to troubleshoot and fix problems with the operating system.
5. True or False: A system restore point is a snapshot of the computer's system files, registry, and configuration settings at a specific point in time, which can be used to revert the system to a previous state if problems occur.

6. True or False: Ping is a command-line utility used to test network connectivity by sending ICMP echo requests to a target device and waiting for ICMP echo replies.

Section 3: Short Answer

7. Describe the steps involved in troubleshooting a computer that fails to boot into the operating system.

Section 4: Practical Application

8. Demonstrate how to troubleshoot network connectivity issues on a Windows computer using the ipconfig command.

Section 5: Essay

9. Discuss the importance of effective communication skills in a helpdesk or technical support role.

Assignment: module -5 Network Fundamentals and Building Networks

Section 1: Multiple Choice

1. What is the primary function of a router in a computer network?
   a) Assigning IP addresses to devices
   b) Providing wireless connectivity to devices
   c) Forwarding data packets between networks
   d) Managing user authentication and access control
2. What is the purpose of DHCP (Dynamic Host Configuration Protocol) in a computer network?
   a) Assigning static IP addresses to devices
   b) Resolving domain names to IP addresses
   c) Managing network traffic and congestion
   d) Dynamically assigning IP addresses to devices
3. Which network device operates at Layer 2 (Data Link Layer) of the OSI model and forwards data packets based on MAC addresses?
   a) Router
   b) Switch
   c) Hub
   d) Repeater
4. Which network topology connects all devices in a linear fashion, with each device connected to a central cable or backbone?
   a) Star
   b) Bus
   c) Ring
   d) Mesh

Section 2: True or

True or False: A VLAN (Virtual Local Area Network) allows network administrators to logically segment a single physical network into multiple virtual networks, each with its own broadcast domain.

True or False: TCP (Transmission Control Protocol) is a connectionless protocol that provides reliable, ordered, and error-checked delivery of data packets over a network.

True or False: A firewall is a hardware or software-based security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.

8. Describe the steps involved in setting up a wireless network for a small office or home office (SOHO) environment.

Section 4: Practical

9. Demonstrate how to configure a router for Internet access using DHCP (Dynamic Host Configuration Protocol).

Section 5:

10. Discuss the importance of network documentation in the context of building and managing networks.

Assignment module 6: Network Security, Maintenance, and Troubleshooting Procedures

Section 1: Multiple Choice

1. What is the primary purpose of a firewall in a network security infrastructure?
   a) Encrypting network traffic
   b) Filtering and controlling network traffic
   c) Assigning IP addresses to devices
   d) Authenticating users for network access
2. What type of attack involves flooding a network with excessive traffic to disrupt normal operation?
   a) Denial of Service (DoS)
   b) Phishing
   c) Spoofing
   d) Man-in-the-Middle (MitM)
3. Which encryption protocol is commonly used to secure wireless network communications?
   a) WEP (Wired Equivalent Privacy)
   b) WPA (Wi-Fi Protected Access)
   c) SSL/TLS (Secure Sockets Layer/Transport Layer Security)
   d) AES (Advanced Encryption Standard)
4. What is the purpose of a VPN (Virtual Private Network) in a network security context?

Section 2: True or false

True or False: Patch management is the process of regularly updating software and firmware to address security vulnerabilities and improve system performance.
True or False: A network administrator should perform regular backups of critical data to prevent data loss in the event of hardware failures, disasters, or security breaches.
True or False: Traceroute is a network diagnostic tool used to identify the route and measure the latency of data packets between a source and destination device.

Section 3: Short

8. Describe the steps involved in conducting a network vulnerability Assignment.

Section 4: Practical Application

9. Demonstrate how to troubleshoot network connectivity issues using the ping command.

Section 5:

10. Discuss the importance of regular network maintenance and the key tasks involved in maintaining network infrastructure.

1. Which of the following best describes the purpose of a VPN (Virtual Private Network)?
   a) Encrypting network traffic to prevent eavesdropping
   b) Connecting multiple LANs (Local Area Networks) over a wide area network (WAN)
   c) Authenticating users and controlling access to network resources
   d) Reducing latency and improving network performance

Assignment:

Module -7: Network fundamental -

1- Which of the following messages in the DHCP process are broadcasted? (Choose two)

A. Request

B. Offer

C. Discover

D. Acknowledge

2- Which command would you use to ensure that an ACL does not block web-based TCP traffic?

A. permit any

B. permit tcp any any eq 80

C. permit tcp any eq 80

D. permit any any eq tcp

3- Explain Network Topologies

4- Explain TCP/IP Networking Model

5- Explain LAN and WAN Network

6- Explain Operation of Switch

7- Describe the purpose and functions of various network devices

7- Make list of the appropriate media, cables, ports, and connectors to 8-

8- connect switches to other

9- Define Network devices and hosts

Module 8: Network Access Basic routing and Advance routing concept, switching concept-

- **Beginner Question**

1. Explain Switch
2. Explain Switch Boot Sequence
3. Explain Three Methods to access Switch Command Line Interface
4. Explain and Configuring the Cisco Internet Operating System
5. Explain Switch Port

4-R1, R2, R3, and R4 have their Fast Ethernet 0/0 interfaces attached to the same VLAN. A network engineer has typed a configuration for each router by using a word processor. He will later copy and paste the configuration into the routers. Examine the following exhibit, which lists configuration for the four routers, as typed by the network engineer. Assuming that all four routers can ping each other's LAN IP addresses after the configuration has been applied, choose the routers that will be able to form a neighbor relationship with the other routers on the LAN. (You can assume that, if not shown in the exhibit, all other related parameters are still set to their defaults.) (Choose two)

```
Configuration (
!
interface faste
   ip address 1
!
router ospf 0
   network 10.(
```

```
Configuration (
!
interface faste
   ip address 1
!
router ospf 1
   network 10.(
```

```
Configuration (
!
interface faste
   ip address 1
!
router ospf 3
   network 10.1
```

```
Configuration (
!
interface faste
   ip address 1
!
router ospf 2
   network 10.(
```

A. R1

B. R2

C. R3

D. R4

E. None of the routers will exchange routing information.

3- enable secret [password] is hashed using the_______algorithm.

A. MD5

B. AH

C. PSK

D. ESP

E. WPA2

4- An engineer connects to Router R1 and issues a show ip ospf neighbor command. The status of neighbor 2.2.2.2 lists FULL/BDR. What does the BDR mean?

A. R1 is an Area Border Router.

B. R1 is a backup designated router.

C. Router 2.2.2.2 is an Area Border Router.

D. Router 2.2.2.2 is a backup designated router.

5- Which command is used to view the neighbor discovery table on a PC?

A. show ipv6 neighbor

B. show ipv6 neighbors

C. netsh interface ipv6 show neighbor

D. netsh interface ipv6 show neighbors

6- What type of variable is being shown? Routers = [R1,R2,R3]

A. List

B. Dictionary

C. Simple

D. Unsigned integers

7- Identify the fields in an IPv4 header. (Choose three)

A. Host component

B. Time to Live

C. Source address

D. Destination address

E. Network address

Assignment: Windows Server

Module 12: Installation, Storage, and Compute with Windows Server

1. What two options are provided in the type of installation window during Windows Server 2016 installation?
2. Write the step How to configure server step by step?
3. What are the Pre installation tasks?
4. What are the Post installation tasks?
5. What is the standard upgrade path for Windows Server?
6. What is the Physical structure of AD?
7. What is the Logical components of Active Directory?
8. What is the Full form Of LDAP?
9. What is the location of the AD database?
10. *What is child DC*?
11. Explain the term forest in AD

12. What is Active Directory? Check all that apply.

- An open-source directory server
- A Windows-only implementation of a directory server
- Microsoft's implementation of a directory server
- An LDAP-compatible directory server

13. When you create an Active Directory domain, what's the name of the default user account?

- Superuser
- Root
- Username
- Administrator

14. AD domain provides which of the following advantages? Check all that apply.

- Centralized authentication
- More detailed logging
- Centralized management with GPOs
- Better performance

15. What are the minimum hardware requirements for installing Windows Server 2016?

16. Explain the different editions of Windows Server 2016 and their features.

17. Walk through the steps of installing Windows Server 2016 using GUI mode.

16. Describe the steps for installing Windows Server 2016 in Server Core mode.

17. How do you configure network settings during Windows Server 2016 installation?

18. Explain the process of promoting a Windows Server to a domain controller.

19. Discuss the steps involved in upgrading from a previous version of Windows Server to Windows Server 2016.

20. What is Active Directory Domain Services (AD DS), and what are its key components?

21. How do you create a new Active Directory user account in Windows Server ?

22. Explain the process of creating and managing Group Policy Objects (GPOs) in Windows Server 2016 or 2019.

23. What are Organizational Units (OUs) in Active Directory, and how do you use them?

24. Describe the process of delegating administrative privileges in Active Directory.

Assignment: Linux

Module: 1 - Linux server - Understand and use essential tools

1. What is the minimum number of partitions you need to install Linux?
2. Explain About Chmod Command
3. How to check Linux memory utilization
4. · Use `grep` to search for specific patterns in files.
5. · Get Connecting on a linux server by ssh
6. · Create 5 files in the /tmp directory, and then use tar and gzip to bundle and compress the files.
7. ·Describe the root account
8. What is shell?
9. What is Linux?
10. What is Bash?
11. You have a new empty hard drive that you will use for Linux. What is the first step you use.
12. Write the Linux command to show the current working directory.
13. write the Linux command to get help with various options.
14. Write the linux comman! to display what all users are currently doing.
15. write the Linux command to get information about the operating system.
16. Write the Linux command to create a hard link of a file.
17. Write the Linux command to create a soft link of a file as well as Directory.
18. Write the Linux command! to search for specific pattern in a file.
19. Write the Linux command to show the use of basic regular expressions using grep command.

Module :2- Linux server - Operate running systems

20. View running processes with `ps`.
21. Terminate processes with `kill`.
22. Use `top` or `htop` to monitor system resources and processes.
23. · Configure one of your lab COMPUTERS to boot to the CLI using systemd, and reboot

to confirm that you were successful.

Module :3- Linux server - Configure local storage Assignment

24. Learn about different filesystem types (e.g., ext4, NTFS).
25. Manage disk partitions and filesystems using tools like `fdisk`, `mkfs`, and `mount`.
26. create a 2048MB partition and verify if the partition has been created.
27. Why LVM is required?
28. How can you find out how much memory Linux is using?
29. What is a typical size for a swap partition under a Linux system?
30. What is the maximum file size on the ext4 file system?
31. What is the maximum file size on the xfs file system

Cloud computing

Module -1
1- What is cloud computing?
2- Describe cloud computing deploy model.
3- What are components of cloud computing?
4- cloud computing advantage and disadvantage Advantages of Cloud Computing

Module -2
1- What is virtualization and virtualization type?
2- Type of hypervisor and how to manage it?
3- Roles of virtualization in cloud computing?
4- What is container?
5- What is high availability and live migration in virtualization?
5- Storage configuration –describe block storage, file storage and object storage---DAS NAS and SAN
6- Describe storage allocation and provisioning. Storage Allocation

Module -3
1- Different type of cloud storage
2- What is role base access control and identity and access management and MFA
3- What is physical and virtual host allocation?
4- How to access resource of cloud computing?
5- Type of backup in cloud?
6- What is disaster recovery?

Module -4
1- Resource Monitoring Techniques
2- How to access compute (windows and Linux) from internet? describe tools and its security
3- Encryption Technologies and Methods
4- Describe network security in cloud, compute security and storage security

Module - 5
1- How to configure, develop and maintain Security and Privacy in cloud?
2- What is Portability in cloud?
3- What is Reliability and high Availability in cloud?
4- Describe Mobility Cloud Computing
5- Describe AWS, Azure, Google cloud Platforms
6- Accessing AWS, Azure and Google cloud Platforms (any one portal)
7- Create compute, create network, create storage on AWS , Azure and GCP
8- Compare Cloud pricing of resources and services on all platform Amazon Web Services (AWS):